

Arithmophysics: What the Horizon Transfer Cost Tells Us to Calculate

Research prompt for Claude, continuing Christophe Michaels's horizon programme

Continue from the corrected Sections 8–9 of SUBPOWER_CLOSURE.md and the paper *Arithmophysics: Prime Modes and Complete Packet Cancellation*. This prompt uses the reported results of those sections; their repository text has not been independently inspected here.

Make a further round trip through the horizon representations. Identify which part of the transfer cost is geometric, which part depends on scale, and which signed arithmetic correlations must be estimated. Attempt an independent inequality for those correlations and carry its quantitative consequences back to the original energy.

An independent estimate can be proved on either side of the horizon, using any rigorously established structure of this framework. “Independent” means that its proof does not assume the desired bound or an equivalent hypothesis. It does not mean that the input must come from outside this framework, or that it must first be a bound on the prime-counting error. Additional laws of the larger model can also be explored; specify the exact step needed to establish them for the arithmetic source.

Treat the established convolution identities and their collapse as checkpoints. Do not spend this pass evaluating them again. The immediate task is an inequality, with a stated range and a stated gain.

1. Retain the exact target and the return map

Write

$$c_N(n) = \mu(n)/n, \quad B_N(a, b) = \min(a, b), \quad b_N = V_N^* c_N,$$

where the columns of V_N and the gains are

$$v_{j,N}(n) = \frac{2}{\sqrt{2N+1}} \sin(n\theta_{j,N}), \quad \theta_{j,N} = \frac{(2j-1)\pi}{2N+1}, \quad \lambda_{j,N} = \frac{1}{4 \sin^2(\theta_{j,N}/2)}.$$

Thus $B_N = V_N \Lambda_N V_N^*$, and

$$R(N) = c_N^* B_N c_N = \sum_{k \leq N} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N}. \quad (1)$$

Keep the terminal term. Set $\ell_N = \log(eN)$ and $J_0(N) = \lceil N/\ell_N^2 \rceil$. For any mode vector z ,

$$K(z) = \max_{1 \leq J \leq N} \lambda_{J,N} \|\Pi_J z\|_2^2, \quad K(z) \leq \|z\|_{\Lambda_N}^2 \leq [1 + 2 \log(3N)] K(z). \quad (2)$$

The original source has an unconditional $O(\ell_N^4)$ high-mode energy bound above J_0 . The low-mode closing scale is

$$\|\Pi_J b_N\|_2^2 \ll_{\varepsilon} J^2 N^{-2+\varepsilon}, \quad 1 \leq J < J_0(N), \quad (3)$$

uniformly in N, J , for every $\varepsilon > 0$. The paper's existing bound is of order $J^2 N^{-1} e^{-c\sqrt{\log N}}$; it is the baseline for comparison, not a claim about the best available classical estimate.

For $K_d = \lfloor N/d \rfloor$, use the inherited dilation $E_d e_m = e_{dm}$ and $T_d = V_N^* E_d V_{K_d}$. With scalar $\Lambda(d)$ denoting von Mangoldt,

$$F_N = \sum_{d=2}^N \frac{\Lambda(d) - 1}{d} T_d b_{K_d}, \quad H_N = V_N^* \text{diag}(\log n) V_N, \quad u_N = V_N^* e_1,$$

$$(H_N - I) b_N = -u_N - F_N, \quad (V_N F_N)_1 = 0, \quad (V_N F_N)_n = -\frac{\mu(n)(\log n - 1)}{n} \quad (n \geq 2). \quad (4)$$

The established uniform inverse gives

$$\sqrt{R(N)} \leq C_*(1 + \sqrt{R^{\log(N)}}), \quad R^{\log(N)} = \|F_N\|_{\mathbf{A}_N}^2. \quad (5)$$

Use a proved uniform C_* ; the supplied multiplier argument gives $C_* < 216$. The reported value near 25.4536 is a finite-dimensional numerical norm unless accompanied by a uniform certification. Do not spend this pass optimizing this fixed constant.

2. Locate the cost before treating it as an arithmetic obstruction

Record the forward and return norms in each representation actually used, including their dependence on N, J and any new cutoff. Separate the contribution of fixed small source indices from a cost that persists at growing scales. A bounded inverse licenses the return trip; it does not itself produce cancellation.

Here is a concrete localization lemma to check and use. For any coefficient vector x supported on $L \leq n \leq N$, with integer $L \geq 3$, let $D_N = \text{diag}(\log n - 1)$. Then

$$\|x\|_{B_N} \leq A_L \|D_N x\|_{B_N}, \quad A_L = \frac{1}{\log L - 1} + \frac{2}{(\log L - 1)^2}. \quad (6)$$

Proof supplied for audit. In the field space $\|f\|_E^2 = \int (|f'|^2 + |f|^2/4) du$, multiplication by a bounded Lipschitz h has norm at most $\|h\|_\infty + 2\|h'\|_\infty$, also on its dual. The source $J_x = \sum \sqrt{n} x_n \delta_{\log n}$ has squared dual norm $x^* B_N x$. Take $h(u) = 1/(\log L - 1)$ for $u \leq \log L$ and $h(u) = 1/(u - 1)$ above it. Multiplication sends $J_{D_N x}$ to J_x and yields (6).

In particular, $A_{21} < 0.968$, uniformly in N . Thus the large full inverse norm is avoidable on a tail beginning at a fixed index. This isolates a finite-index conditioning effect; it supplies no power saving in the arithmetic energy. A tail restriction is not an orthogonal projection in the B_N metric: account for the removed source and its cross terms. If L grows with N , track that dependence explicitly.

Do not commute Π_J with H_N , its inverse, or a source cutoff without estimating the resulting error.

3. A concrete calculation: the diagonal already has the required scale

Use either of the two real weights

$$w_n = 1, \quad \text{or} \quad w_1 = 0, \quad w_n = \log n - 1 \quad (n \geq 2).$$

Define

$$z_N^{(w)} = V_N^*(\mu(n)w_n/n)_{n \leq N}, \quad \mathcal{K}_{N,J}(a, b) = \sum_{j \leq J} v_{j,N}(a)v_{j,N}(b).$$

The second choice gives $z_N^{(w)} = -F_N$. Decompose exactly:

$$\begin{aligned} \|\Pi_J z_N^{(w)}\|_2^2 &= \mathcal{D}_{N,J}^{(w)} + \mathcal{O}_{N,J}^{(w)}, \\ \mathcal{D}_{N,J}^{(w)} &= \sum_{n \leq N} \frac{\mu(n)^2 w_n^2}{n^2} \mathcal{K}_{N,J}(n, n), \\ \mathcal{O}_{N,J}^{(w)} &= 2 \sum_{1 \leq a < b \leq N} \frac{\mu(a)\mu(b)w_a w_b}{ab} \mathcal{K}_{N,J}(a, b). \end{aligned} \quad (7)$$

For the unweighted diagonal, there is an unconditional bound

$$\boxed{\mathcal{D}_{N,J}^{(1)} \leq \frac{2\pi^2 J^2}{(2N+1)^2} \leq \frac{\pi^2 J^2}{2 N^2}.} \quad (8)$$

Indeed, $\mu(n)^2 \leq 1$, and the classical Fourier identity

$$\sum_{n \geq 1} \frac{\sin^2(n\theta)}{n^2} = \frac{\pi\theta - \theta^2}{2} \quad (0 \leq \theta \leq \pi)$$

gives a mode- j contribution at most $2\pi^2(2j-1)/(2N+1)^2$. Sum over $j \leq J$. For the log weight,

$$\mathcal{D}_{N,J}^{(w)} \leq \ell_N^2 \frac{2\pi^2 J^2}{(2N+1)^2}. \quad (9)$$

Equations (8)–(9) are elementary diagonal estimates, not estimates of the arithmetic cross terms. They tell us exactly where the remaining work lies.

Primary inequality to attempt: for at least one weight, prove

$$\boxed{\mathcal{O}_{N,J}^{(w)} \leq C_\varepsilon J^2 N^{-2+\varepsilon}, \quad 1 \leq J < J_0(N),} \quad (10)$$

with the stated uniformity. This is an upper bound on the combined signed contribution. A separate absolute-value bound on every lag or every horizon pair is not required. An intermediate improvement on a specified growing range is also a useful result; state exactly which ranges remain uncontrolled.

4. Make the excursion through the horizon useful

Start with the horizon organization of the log-weighted case. Group the exact dilations into dyadic blocks:

$$q_{r,N} = \sum_{\substack{2^r \leq d < 2^{r+1} \\ 2 \leq d \leq N}} \frac{\Lambda(d) - 1}{d} T_d b_{\lfloor N/d \rfloor}, \quad r \geq 1.$$

Keep the full projected Gram form

$$\|\Pi_J F_N\|_2^2 = \sum_{r,s} \operatorname{Re} \langle \Pi_J q_{r,N}, \Pi_J q_{s,N} \rangle. \quad (11)$$

All integer dilations belong here, including composites on which $\Lambda(d) - 1 = -1$. Preserve the quotient cutoffs, aliases, floor errors, and correlations between the actual child vectors. The block diagonal in (11) is different from the source diagonal in (7); do not identify them.

Determine whether the exact transport reveals a family of cross-block terms that can be bounded together using the arithmetic of the source. If so, derive the inequality and translate it into (10). If a proposed estimate is valid for arbitrary child vectors, test its worst case before using it for the correlated arithmetic family.

For a return to arithmetic coordinates, the equivalent lag sum is

$$\mathcal{O}_{N,J}^{(w)} = 2 \sum_{h=1}^{N-1} \sum_{a=1}^{N-h} \frac{\mu(a)\mu(a+h)w_a w_{a+h}}{a(a+h)} \mathcal{K}_{N,J}(a, a+h). \quad (12)$$

Its exact kernel is available without approximation. Put $D = 2N + 1$ and $H_J(t) = \sin(2Jt)/\sin t$, using $H_J(0) = 2J$. Then

$$\mathcal{K}_{N,J}(a, b) = \frac{1}{D} \left[H_J\left(\frac{\pi(a-b)}{D}\right) - H_J\left(\frac{\pi(a+b)}{D}\right) \right]. \quad (13)$$

Attempt an estimate for these actual weights. If using a known averaged correlation or bilinear-sum theorem, check its averaging measure, exceptional sets, endpoints, uniformity in J , and resulting error after summation. An average over horizons needs a proved passage to every horizon. A smooth-density approximation needs a discrepancy estimate in the growing-mode range where it is used.

One possible closing mechanism is an independently derived recurrence

$$\begin{aligned} R(N) &\leq \sum_{k < N} a_{N,k} R(k) + E_N, & a_{N,k} &\geq 0, \\ \sum_{k < N} a_{N,k} (k/N)^\varepsilon &\leq 1 - \delta_\varepsilon, & E_N &\leq C_\varepsilon N^\varepsilon, & \delta_\varepsilon &> 0. \end{aligned} \quad (14)$$

These conditions would allow strong induction after treating the finite initial range. They are a proposed sufficient mechanism, not established properties of the transfers. Derive any such coefficients and error from an actual estimate; do not insert the target bound for the child horizons in place of the missing argument. A different rigorous mechanism is equally acceptable.

5. Compute diagnostics that can discriminate between mechanisms

Use a feasible geometric sequence of N , with low-mode cutoffs ranging from fixed J to just below $J_0(N)$. Report the achieved range. Compute the normalized quantities

$$\frac{N^2}{J^2} \mathcal{D}_{N,J}^{(w)}, \quad \frac{N^2}{J^2} \mathcal{O}_{N,J}^{(w)}, \quad \frac{N^2}{J^2} \|\Pi_J z_N^{(w)}\|_2^2.$$

For moderate sizes, also show the signed Gram contributions in (11), grouped by dilation scale. Identify which interactions create positive excess and which suppress it. Check small cases by direct summation before using fast transforms. Use adequate precision when recovering a small difference between larger terms.

Compare the actual source with two controls on the same squarefree support: independent random signs and all positive signs. For independent centered signs, the expected off-diagonal contribution in (7) is zero; this is a control calculation, not an assumption about Möbius. A proposed geometric estimate that cannot distinguish the coherent control has not yet used the needed arithmetic information.

The newly supplied kernel and diagonal bound were checked here at $N = 16, 31, 128, 1000, 4000$, and the tail multiplier bound against finite matrix norms. These checks validate implementation examples, not the uniform arithmetic inequality (10). Concentrate new computation on testing a specific proposed estimate, rather than fitting an exponent to a short table.

6. Bring the result back and state precisely what was gained

For the log-weighted route,

$$\|F_N\|_2^2 \leq C_{\log} := \sum_{n \geq 2} \frac{(\log n - 1)^2}{n^2} < \infty.$$

Consequently the modes above J_0 contribute at most $(9/4)C_{\log}\ell_N^4$ in the energy metric. A uniform proof of (10), combined with (9), (2), and this high-mode bound, gives subpower $R^{\log}(N)$; equation (5)

returns that estimate to $R(N)$. Use a smaller positive exponent before absorbing logarithms. For the unweighted route, use (8) and return directly through (2). No separate proof of both routes is needed.

Deliver a proposed next research section containing:

1. The strongest new inequality actually proved, its quantifiers, and its independent input. If no arithmetic improvement is proved, say so once and identify the exact attempted inequality and the obstruction found.
2. The calculation showing what each horizon transfer costs and what the round trip preserves. Include the finite-index localization and any scale-dependent losses.
3. Reproducible diagnostics that support or refute a specific mechanism, with the corresponding signed correlations identified.
4. The return calculation, including all omitted-range and terminal terms, and the sharpest remaining mathematical question.

Keep the corrected completion formula for every $P \geq \sqrt{N}$,

$$M_{Q(P)}(N) = M(N) + \sum_{P < p \leq N} M(\lfloor N/p \rfloor).$$

The prime-count simplification applies only when $P \geq N/2$. Retain the earlier counterexamples to a uniform per-prime contraction and to a generic positive log-weight anticommutator; any new claim of that kind must address those examples.

The purpose of the round trip is to discover an estimate made accessible by the other representation, and then return with its gain intact. The representation may tell us where to calculate; the calculation must determine whether a gain exists.

Reference anchors

- NIST DLMF, §27.5, [Inversion Formulas](#), for the classical convolution background.
- NIST DLMF, [27.11.12](#), for the classical Möbius partial-sum bound underlying the stated baseline.
- NIST DLMF, [24.8.1](#), whose B_2 Fourier series yields the sine-square identity used in (8).

The horizon formulas are the supplied research checkpoint. Equations (6) and (8)–(9) are additional elementary deductions supplied with this prompt. Equation (10) is a research target, not a proved conclusion.